

Noisy calibration, latent coverage and the location of the mean

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SUMMARY

Prediction intervals for a latent quantity, such as the mean of an unsampled area, are often calibrated on noisy proxies with Gaussian errors. A symmetric unimodal latent distribution makes such calibration conservative, but symmetry is seldom plausible. For log-concave and, more generally, bi-log-concave latent distributions we show that the correction to the calibrated radius needed in the worst case is governed by the distance between the latent mean and the boundary of the interval. If that distance is bounded away from zero, the correction is at most of the order of the noise variance at every noise level, and asymmetric distributions with mean zero attain this order. Without a restriction on the mean the correction is of the order of the noise standard deviation, with an explicit sharp constant; a correction of that order requires the mean to lie within a bounded multiple of the noise standard deviation of the boundary, and the limiting coefficient is then a one-dimensional extremum. A calibration slack of one part in a thousand at the ninety per cent level removes the correction at every noise level. As a consequence, the noisy conformal threshold at a suitable rank covers the latent target with the nominal reliability even when the noise variances are heterogeneous and unknown, and known variances shorten it by about ten per cent in our simulations.

Some key words: Conformal prediction; Deconvolution; Log-concavity; Measurement error; Small area estimation.

1. INTRODUCTION

Suppose that a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$, where $e \sim N(0, D)$ is independent of W and D is known. A survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas is in this position (Fay & Herriot, 1979), as is a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t at which the noisy scores $|V|$ have coverage p is observable. We ask which radius gives the latent residual coverage q , uniformly over a class of latent distributions.

If W is symmetric and unimodal, the theorem of Anderson (1955) gives $\text{pr}(|W + e| \leq t) \leq \text{pr}(|W| \leq t)$, so the noisy threshold is conservative (Einbinder et al., 2024). Without symmetry the available transfers make no shape assumption: a median-zero noise turns a noisy level $1 - \alpha$ into a latent level $1 - 2\alpha$, with high-probability versions under a Beta calibration (Guille-Escuret & Ndiaye, 2024), and LatentCP (Zheng et al., 2026) uses a Markov-type compatibility threshold. Cohen et al. (2025) estimate the noise-free threshold by deconvolution, without a latent guarantee, and Armstrong et al. (2022) give worst-case empirical Bayes intervals for sampled units. For a fixed smooth latent distribution, Gaussian noise moves quantiles by $O(D)$ (Jochmans & Weidner, 2024), which suggests that the error of noisy calibration is of second order.

Uniformly over a shape class the picture is different, and it is organized by the location of the mean. We work with log-concave laws and with the larger class of bi-log-concave laws (Dümbgen et al., 2017), which contains multimodal ones. We show that without a restriction on the mean the worst-case radius correction at equal levels is of order $D^{1/2}$, with a sharp constant (Theorem 1); that if the mean lies at distance d from the boundary of the interval the correction is at most $D/(2d)$ for every D , an order attained by asymmetric laws with mean zero (Theorem 2, Proposition 2); and that a correction of order $D^{1/2}$ requires the mean within $O(D^{1/2})$ of the boundary, with a limiting coefficient given by a one-dimensional extremum (Theorem 3). These are worst-case statements over a class; a particular law with its mean near the boundary need not need any correction. A calibration slack of order 10^{-3} removes the correction at every noise level, so the noisy threshold at a suitable rank is valid for the latent target even when the noise variances are heterogeneous and unknown (Proposition 4); known variances then serve efficiency (§ 4).

2. THE SHARP RADIUS AND ITS REDUCTION

Let $\mathcal{L}$ be the class of log-concave laws on the real line, including point masses, and $\mathcal{B} \supset \mathcal{L}$ the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave. Both classes are closed under translation and scaling, and convolution with a log-concave law preserves them (Saumard, 2019). After scaling t to 1, write $x = D/t^2$ and let $Z \sim N(0, 1)$ be independent of W . Define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q is the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible, which happens exactly when $\text{pr}(|x^{1/2}Z| \leq 1) < p$. If calibration certifies noisy coverage at least p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law. This is sharpness within the class and for one constraint, not optimality among procedures that use the whole calibration sample. We call $R_{p,q}(x) - 1$ the radius correction; it is a correction to a length, not a coverage probability. For $0 \leq b < 1$ let $R_{p,q}^{(b)}(x)$ be defined by (1) with the additional constraint $|E(W)| \leq b$, so that $1 - b$ bounds the distance of the mean from the boundary.

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1) = \Phi\{(1 - w)/x^{1/2}\} - \Phi\{(-1 - w)/x^{1/2}\}$, so that the constraint in (1) reads $E\{g_x(W)\} \geq p$. Let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment $[a, b]$.

PROPOSITION 1. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § 4.2.

The proof, in the Supplementary Material, truncates W to an interval outside which the kernel is below p , which keeps the constraint, and applies the localization theorem of Fradelizi & Guédon (2004). For these laws the constraint is a combination of normal distribution functions, and log-concavity in the translation (Prékopa, 1973) reduces the supremum in (1) to a two-dimensional maximization of closed-form quantities.

3. WORST-CASE RADIUS CORRECTION AND THE LOCATION OF THE MEAN

3.1. A one-sided problem

Near each end of the interval the problem is one-sided. For $u > 0$ let E_u be exponential with rate u , independent of Z , and for the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub+u^2/2} \Phi(b - u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$. The law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$. 80

LEMMA 1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing.

LEMMA 2. Let $e^{-1} < q < 1$ and $\kappa \geq 0$. Then

$$\sup\{F_Y^{-1}(q) : Y \in \mathcal{B}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\}.$$

The function L_q is positive and nonincreasing, $L_q(\kappa) = c_q$ for $\kappa \leq \kappa_q^*$, where κ_q^* is the largest value of k_q at a maximizer of v_q , and $\kappa L_q(\kappa) \rightarrow \{1 - \log(1/q)\}/2$ as $\kappa \rightarrow \infty$. 85

Both proofs replace Y by the exponential tail through a supergradient of $\log F_Y$ at its q -quantile; the replacement lies stochastically below Y , so it keeps the quantile, the noisy constraint and the bound on the mean. This uses only concavity of $\log F_Y$, so Lemma 1 holds over $\mathcal{B}$. Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$, and in double precision $\kappa_q^* = 6.81, 20.18$ and 50.07 (Supplementary Material). 90

3.2. No restriction on the mean

THEOREM 1. For every $q \in (0, 1)$ and every $x > 0$, $R_{q,q}(x) \leq 1 + c_q x^{1/2}$, and the bound holds for every $W \in \mathcal{B}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$.

Proof of the upper bound. Let $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, and let $\alpha_R = \text{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \text{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave and $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R$. By Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$, which uses concavity of $\log(1 - F_W)$, gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$, so $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. $\square$ 95

The lower bound, proved in the Supplementary Material, takes $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ at a near-optimal rate u , with $\eta_x \rightarrow 0$. Its density rises to a cut-off just outside the threshold, noise draws more mass inwards across that end than it pushes out, and its mean, $1 - k_q(u)x^{1/2}$, is within $O(x^{1/2})$ of the boundary; the next results show that this is necessary. 100

3.3. The mean away from the boundary

THEOREM 2. Let $W \in \mathcal{B}$ with $|E(W)| \leq B$, let $\sigma > 0$ and $t > B$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then 105

$$Q_q(|W|) \leq B + \{(t - B)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - B)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$, and $Q_q(|W|) \leq (t^2 + \sigma^2)^{1/2}$ if $E(W) = 0$.

The proof in the Appendix bounds the score of the smoothed law by comparison with an exponential tail and uses the heat equation for quantiles (Steinbrecher & Shaw, 2008) to obtain $d\{r(s) - B\}^2/ds \geq -1$ for $r(s) = Q_q(|W + s^{1/2}Z|)$. Unlike the expansion for a fixed smooth law, the bound is uniform over the class. The order x cannot be improved, even for mean-zero laws. 110

PROPOSITION 2. Let X have density proportional to $e^{\beta X}$ on $[-\ell, 0]$ with $\beta > 0$ and $0 < \ell \leq \infty$, let $W = X - E(X)$ and let $r_0 = Q_q(|W|)$. If $\pm r_0$ lie in the interior of the support of W , then, after

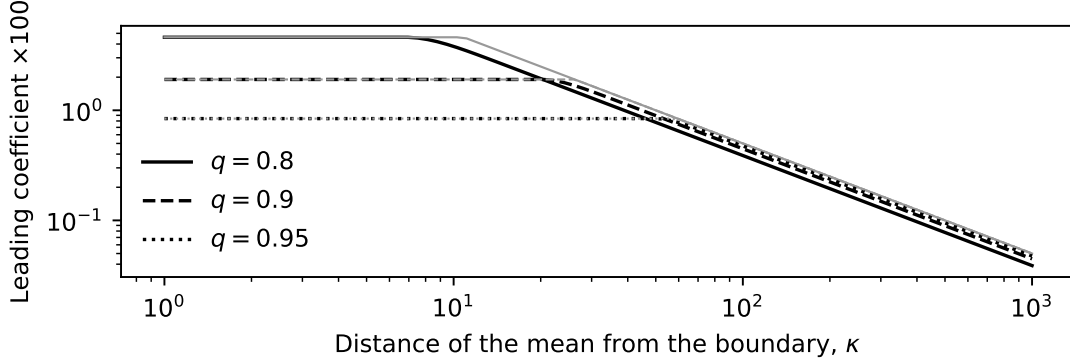


Fig. 1. Leading coefficient $L_q(\kappa)$ of the worst-case radius correction, times 100, when the mean lies at distance at least $\kappa x^{1/2}$ from the boundary (Theorem 3), for $q = 0.8$ (solid), 0.9 (dashes) and 0.95 (dots), with the upper bound $\min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ from Theorems 1 and 2 (grey). Both axes are logarithmic.

Alt text: Line plot on logarithmic axes of the leading coefficient against the scaled distance of the mean from the boundary for three coverage levels. Each curve is flat at its unrestricted constant up to a threshold near 7, 20 and 50 and then decreases in proportion to the reciprocal of the distance, slightly below a grey upper bound.

scaling W so that the noisy q -quantile of $|W + x^{1/2}Z|$ equals 1,

$$Q_q(|W|) = 1 + \frac{1}{2}\beta r_0 \tanh(\beta r_0) x + o(x) \quad (x \rightarrow 0).$$

115 For every $q \in (0, 1)$ the condition holds for small $\beta\ell$, so $R_{q,q}^{(0)}(x) - 1$ is of exact order x .

The best coefficient in this family is 0.0747 at $q = 0.9$ and 0.347 at $q = 0.8$, for $W = 1 - E_1$, so the worst-case coefficient over centred laws lies in $[0.0747, 0.5]$ and $[0.347, 0.5]$. Larger coefficients come with quantiles closer to the end of the support, and the correction appears only for noise much smaller than that gap.

120 3.4. The transition

THEOREM 3. Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\delta = Q_q(|W|) - 1 > 0$,

$$2\{1 - |E(W)|\}\delta + \delta^2 \leq x. \quad (2)$$

Moreover, for every $\kappa \geq 0$,

$$\lim_{x \rightarrow 0} x^{-1/2} \{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa), \quad b_x = 1 - \kappa x^{1/2},$$

and the same limit holds with $\mathcal{L}$ replaced by $\mathcal{B}$ in the definition of $R^{(b)}$.

125 By (2), a correction $\delta \geq ax^{1/2}$ with $0 < a < 1$ forces $1 - |E(W)| \leq (1 - a^2)x^{1/2}/(2a)$, and the limit identifies the coefficient on that scale. In the original units the mean lies $\kappa D^{1/2}$ from the threshold, and the unrestricted worst case $c_q D^{1/2}$ persists until κ exceeds κ_q^* , about 20 noise standard deviations at $q = 0.9$; beyond it the coefficient decays like $\{1 - \log(1/q)\}/(2\kappa)$ (Fig. 1). The limit, with κ fixed, does not give the centred coefficient.

4. CALIBRATION SLACK AND A FINITE-SAMPLE PROCEDURE

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4.1. Slack removes the correction

The proof of Theorem 1 splits the noisy failure budget between the two ends, and it extends to $p \geq q$:

$$R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}, \quad C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}),$$

for every x . Numerically the worst split puts the whole budget on one end, so that $C_{p,q} = c_{p,q}$.

COROLLARY 1. *If $C_{p,q} \leq 0$ then $R_{p,q}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x . The smallest slack $p - q$ with $C_{p,q} \leq 0$ lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.8, 0.9$ and 0.95 , and so does the smallest slack with $R_{q+\delta,q}(x) \leq 1$ for all x .*

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The bound on $R_{p,q}$ and Corollary 1 hold for every $W \in \mathcal{B}$, since the proof of Theorem 1 uses only concavity of $\log F$ and $\log(1 - F)$. Ball arithmetic also gives $-0.0572 \leq C_{0.9036, 0.9} \leq -0.057$ and $C_{0.9068, 0.9} \leq -0.114$.

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4.2. Heterogeneous and unknown noise variances

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law G , independent of $e_i \sim N(0, D_i)$ with the D_i fixed. The scores $V_i = W_i + e_i$ are independent but not identically distributed, and $\mathcal{D}$ denotes the calibration data.

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PROPOSITION 3. *Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the Beta($k, K + 1 - k$) distribution. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

The proof, by Hoeffding (1956), is in the Supplementary Material; the result is a case of the comparison of Sen (1970).

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PROPOSITION 4. *Let $G \in \mathcal{B}$, let the D_i be unknown, and suppose that $k \geq Kp_k + 1$ and $C_{p_k, q} \leq 0$. Then $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq T \mid \mathcal{D}) \geq q\} \geq 1 - \delta$.*

Proof. On the event of Proposition 3, $K^{-1} \sum_i \text{pr}(|W + e_i| \leq T) \geq p_k$ for $W \sim G$. If $\text{pr}(|W| \leq T) < q$, Corollary 1 at threshold T gives $\text{pr}(|W + e_i| \leq T) < p_k$ for every i , whatever D_i , and so for the average, a contradiction. $\square$

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At $q = 0.9$, $C_{p, 0.9} \leq 0$ for $p \geq 0.901$, and $K = 110$, $k = 105$, $\delta = 0.05$ give $p_k = 0.9068$: the noisy threshold is valid for the latent target with no knowledge of the variances. Known variances shorten it.

COROLLARY 2. *Under the conditions of Proposition 4, if $D_i \geq D_{\min}$ for all i and $C_{p', q} < 0$ for some $p' \leq p_k$, then $s = \max\{0, T + C_{p', q}D_{\min}^{1/2}\}$ has the same guarantee.*

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Proof. On the event of Proposition 3, some i has $\text{pr}(|W + e_i| \leq T) \geq p_k \geq p'$. The bound $R_{p', q}(x) \leq 1 + C_{p', q}x^{1/2}$ at $x = D_i/T^2$ gives $Q_q(|W|) \leq T + C_{p', q}D_i^{1/2} \leq T + C_{p', q}D_{\min}^{1/2}$. $\square$

At $q = 0.9$ this gives $s = T - 0.114D_{\min}^{1/2}$; it uses one negative constant, not the smallest-variance kernel, which is not a valid plug-in (Supplementary Material). The sharpest use of known variances is the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$: for log-concave G , $\int \bar{g} dG \geq p_k$ on the event of Proposition 3, Proposition 1 applies, and $s = TR_{p_k - \varepsilon, q}^{\text{mix}}(D_1/T^2, \dots, D_K/T^2)$ has the same guarantee, ε bounding a discretization error. We compute R^{mix} by branch and bound

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Table 1. *Reliability, times 100, and mean width, in parentheses; target reliability 95*

latent law	Markov	Fay–Herriot	noisy threshold	Corollary 2	computed
normal	100 (5.92)	96 (3.93)	100 (4.96)	100 (4.90)	98 (4.45)
Laplace	100 (6.29)	93 (3.96)	100 (5.15)	100 (5.09)	99 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.90)	100 (4.45)
truncated Laplace	100 (5.89)	85 (3.89)	100 (4.93)	100 (4.87)	97 (4.42)

Noisy threshold, T at rank 105; computed, TR^{mix} ; centred gamma with shape 2. Standard errors are at most 1.8 for reliabilities and 0.07 for widths.

with double-precision closed forms and a margin of 10^{-9} , not interval arithmetic, whereas Proposition 4 and Corollary 2 rest only on ball-arithmetic constants.

5. NUMERICAL ILLUSTRATION

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$, $\text{var}(W) = 1$ and heterogeneous noise variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Conditional coverage is exact, from closed-form latent distribution functions, with 150 data sets per law. We compare the three rules of § 4.2; the Fay–Herriot tolerance interval with a parametric bootstrap (Chatterjee et al., 2008); and the Markov rule applied to $\bar{g}_T(0) - \bar{g}_T(W) \geq 0$, the smallest radius the single constraint allows without a shape assumption, at rank 107, the best of 106–110 here.

The three rules with a latent guarantee reached the target for every law. Corollary 2 shortens the noisy threshold by 1.2–1.3% and the computed radius by 9.5–10.5%, which is 25–28% shorter than the Markov rule. The Fay–Herriot interval is 12.5–18% shorter but misses the target under the truncated Laplace law. With widths rescaled by an oracle to equal reliability, noisy calibration is not longer, so the gains are in the cost of a guarantee. LatentCP (Zheng et al., 2026), with its marginal guarantee and the new area’s noise variance drawn from the generating distribution, was 16–18% longer than the computed radius, and the deconvolution threshold of Cohen et al. (2025) had mean coverage 0.88–0.90 and reliability 0.45–0.67 in a stabilized version (Supplementary Material).

6. DISCUSSION

The guarantees are conditional on the calibration sample and concern a new area whose residual is drawn from the law of the calibration residuals, not each fixed area or a fixed finite population; they assume independent Gaussian noise. The mean condition of Theorems 2 and 3 concerns the latent residual law, which an intercept in the fitted regression does not centre.

Open questions include the leading coefficient for centred laws, between 0.0747 and 0.5 at $q = 0.9$, a median version of Theorem 2, which Gaussian smoothing does not preserve, and a proof that $C_{p,q} = c_{p,q}$.

DECLARATION OF THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

[To be completed by the author.]

SUPPLEMENTARY MATERIAL

The Supplementary Material includes omitted proofs, the certificates, the estimated-variance procedure, further simulations and an application. Code reproducing every number is at <https://github.com/kunhailP/noisy-calibration-latent-coverage>.

APPENDIX 1

Proofs for § 3

The remaining proofs are in the Supplementary Material.

Proof of Lemma 2. Let $a = \log(1/q)$ and $A = 1 - a > 0$. Let Y be nondegenerate with q -quantile v . Since $\log F_Y$ is concave, a supergradient $u > 0$ at v gives $F_Y(y) \leq \min\{qe^{u(y-v)}, 1\}$, the distribution function of $Y' = v + a/u - E_u$; so Y' lies stochastically below Y and has the same q -quantile. Then $E(Y') \leq E(Y) \leq -\kappa$ and $\Pr(Y' + Z \leq 0) \geq q$, so the replacement keeps both constraints. For the tail $b - E_u$ the constraints read $b \leq b_u(q)$ and $b \leq u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A/u - \kappa\}$. If $k_q(u) < \kappa$ this minimum is $A/u - \kappa$; since k_q is continuous and $k_q(s) \rightarrow \infty$ as $s \rightarrow 0$, some $s < u$ has $k_q(s) = \kappa$, where the value is $A/s - \kappa > A/u - \kappa$. So only u with $k_q(u) \geq \kappa$ matter, and there the value is $v_q(u)$. A point mass y_0 satisfying the constraints has $y_0 = E(Y) \leq -\kappa \leq 0 < L_q(\kappa)$, where positivity holds because $v_q(u) \sim u/2$ and $k_q(u) \sim A/u$ as $u \rightarrow 0$. The same expansions give $\kappa L_q(\kappa) \rightarrow A/2$: for large κ only small u are feasible, $u \leq \{A + o(1)\}/\kappa$, and $v_q(u) \leq \{1 + o(1)\}u/2$ with equality at the boundary. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. $\square$

Proof of Theorem 2. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq B$. Otherwise, for $s > 0$ the law of $W + s^{1/2}Z$ is in $\mathcal{B}$ (Saumard, 2019), with mean μ , distribution function F_s and a positive smooth density f_s . For $r > \mu$ let $h = f_s(r)/F_s(r)$. The tangent of $\log F_s$ at r gives $F_s(y) \leq \min\{F_s(r)e^{h(y-r)}, 1\}$, the distribution function of an exponential tail with mean $r - \{1 + \log F_s(r)\}/h$ that lies stochastically below $W + s^{1/2}Z$, so $\mu \geq r - \{1 + \log F_s(r)\}/h$ and $h(r - \mu) \leq 1$. Laws in $\mathcal{B}$ satisfy $f'_s \leq f_s^2/F_s$ (Dümbgen et al., 2017), so $(\log f_s)'(r) \leq h \leq 1/(r - \mu)$; applied to $-W$ this gives $(\log f_s)'(-r) \geq -1/(r + \mu)$. Hence, for $r > B$,

$$f'_s(r) \leq \frac{f_s(r)}{r - B}, \quad -f'_s(-r) \leq \frac{f_s(-r)}{r - B}.$$

Let $H_s(r) = \Pr(|W + s^{1/2}Z| \leq r)$. By the heat equation $\partial_s f_s = f''_s/2$,

$$\partial_s H_s(r) = \frac{1}{2} \{f'_s(r) - f'_s(-r)\} \leq \frac{f_s(r) + f_s(-r)}{2(r - B)} = \frac{\partial_r H_s(r)}{2(r - B)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \geq -1/[2\{r(s) - B\}]$ where $r(s) > B$. Hence $\{r(s) - B\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\epsilon) - B\}_+^2 \leq \{r(\sigma^2) - B\}_+^2 + \sigma^2 - \epsilon$. As $\epsilon \rightarrow 0$, $r(\epsilon) \rightarrow Q_q(|W|)$, because a nondegenerate law in $\mathcal{B}$ has a positive density on the interior of its support (Dümbgen et al., 2017), so the distribution function of $|W|$ is continuous and strictly increasing where it lies strictly between 0 and 1. Since $r(\sigma^2) \leq t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \leq d + \sigma^2/(2d)$. $\square$

Proof of Proposition 2. Near $\pm r_0$ the density f of W is smooth with $f' = \beta f$, and the ends of the support are at positive distance. Hence $H_v(r) = \Pr(|W + v^{1/2}Z| \leq r) = H_0(r) + v\{f'(r) - f'(-r)\}/2 + o(v)$ uniformly near r_0 , and the noisy quantile is $t(v) = r_0 - v\beta\{f(r_0) - f(-r_0)\}/[2\{f(r_0) + f(-r_0)\}] + o(v) = r_0 - (\beta v/2) \tanh(\beta r_0) + o(v)$. Scaling by $t(v)$ and writing $x = v/t(v)^2$, a continuous and increasing function of v near 0, gives the expansion. As $\beta\ell \rightarrow 0$, W/ℓ tends to the uniform law on $[-1/2, 1/2]$, whose q -quantile of $|W|/\ell$ is $q/2 < 1/2$. $\square$

Proof of Theorem 3. Let $V = W + x^{1/2}Z$ with distribution function F . Since $\log F$ and $\log(1 - F)$ are concave and $F(V)$ is uniform, $E\{\log F(V)\} = -1$ and Jensen's inequality gives $F\{E(V)\} \geq e^{-1}$ and likewise $1 - F\{E(V)\} \geq e^{-1}$. If $E(W) \geq 1$ then $\Pr(|V| \leq 1) \leq F\{E(W)\} \leq 1 - e^{-1} < q$, and similarly

if $E(W) \leq -1$; so $|E(W)| < 1$. Theorem 2 with $B = |E(W)|$ and $t = 1$ gives $\delta \leq \{d^2 + x\}^{1/2} - d$ with $d = 1 - |E(W)|$, which is (2).

For the lower bound in the limit, take u with $k_q(u) \geq \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ of the lower bound in Theorem 1. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \leq 1 - \kappa x^{1/2}$, because b_u decreases in its argument, and it is positive for small x ; hence $\liminf_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \geq v_q(u)$ for every such u .

For the upper bound, suppose that $x_n \rightarrow 0$, $h_n = x_n^{1/2}$, and that W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$ with $Q_q(|W_n|) - 1 \geq a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (2), $\kappa h_n \leq 1 - |E(W_n)| \leq K h_n$ with $K = (1 - a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \geq 0$; let $Y_n = (W_n - 1)/h_n$, so that $-K \leq E(Y_n) \leq -\kappa$. The law of $T_n = Y_n + Z$ is in $\mathcal{B}$, with a smooth positive density of maximum M_n and distribution function F_n , and $F_n(0) \geq \Pr(|W_n + h_n Z| \leq 1) \geq q$ while $F_n\{E(T_n)\} \leq 1 - e^{-1}$ by the first part. So $q - 1 + e^{-1} \leq F_n(0) - F_n\{E(T_n)\} \leq M_n K$. For a law in $\mathcal{B}$ with smooth density g , maximum M and median m , the hazard $g/(1 - F)$ is nondecreasing, so $g(y) \leq 2g(m)\{1 - F(y)\} \leq 2g(m)$ for $y \leq m$, and g/F is nonincreasing, which gives the same for $y \geq m$; so $g(m) \geq M/2$; the tangent bounds for $\log F$ and $\log(1 - F)$ at m give $\Pr(|T - m| > y) \leq e^{-2g(m)y}$, hence $\text{var}(T) \leq 1/\{2g(m)^2\} \leq 2/M^2$. Thus $\text{var}(Y_n) \leq 2K^2/(q - 1 + e^{-1})^2$, the Y_n are tight and uniformly integrable, and along a subsequence $Y_n \rightarrow Y$ in distribution with $Y \in \mathcal{B}$, possibly a point mass, since $\mathcal{B}$ is closed under weak convergence (Saumard, 2019), $E(Y) \leq -\kappa$ and $\Pr(Y + Z \leq 0) = E\{\Phi(-Y)\} \geq q$. On $\{W_n \geq 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so $a_0 \leq \{Q_q(|W_n|) - 1\}/h_n \rightarrow Q_q(Y) \leq L_q(\kappa)$ by Lemma 2, where the quantile converges because a law in $\mathcal{B}$ has a unique q -quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$, so $\limsup_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \leq a_0$ for every $a_0 \in (L_q(\kappa), 1)$; if $L_q(\kappa) \geq 1$ there is nothing to prove, because (2) gives $\delta < x^{1/2}$. $\square$

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